



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

Semester II 2025/2026

Subject : PROBABILITY & STATISTICAL DATA ANALYSIS (SECI 1143)
Task : Chapter 5.3 & 6
Due Date : **Week 14 (15 – 21 June 2026)**

INSTRUCTION:

- i. This is a **GROUP** assignment (each group 3 – 4 members only). Please clearly write the group members **NAME & MATRIC NUMBER** in the front page of the submission.
 - ii. This assignment contributes to 5% of overall course marks.
 - iii. Only **HANDWRITTEN** submission is accepted:
 - a. Submissions using any reporting or statistical tools (e.g.: MS Word, MS Excel, etc.,) will be **REJECTED**.
 - b. Make sure the submission is neatly written. Any submission with handwriting that is unreadable, will be **REJECTED**.
 - c. For answer that need to draw graphs, using graph paper is optional. You can use plain paper.
 - d. Round your **final answers** to **THREE** decimal places.
 - e. Please scan/snapshot your work and save as a PDF file.
 - iv. Submission via eLearning – only **ONE** group member needs to submit on behalf of the group.
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QUESTION 1**(10 MARKS)**

A hospital researcher wants to determine whether vaccinated and non-vaccinated individuals differ in their willingness to participate in a free health screening programme.

A random sample of individuals was selected and the results are shown below:

Group	Number of Individuals	Agreed to Participate
Vaccinated	150	105
Non-vaccinated	120	66

Using $\alpha = 0.05$, test whether there is a significant difference between the proportions of vaccinated and non-vaccinated individuals who agreed to participate.

QUESTION 2**(10 MARKS)**

A manufacturing company wants to determine whether the variability in packaging weights produced by a new automated machine is smaller than the variability produced by the old machine.

Two independent samples are selected, and the results are shown below:

Machine	Sample Size	Sample Variance
New Machine	14	9
Old Machine	12	20

Assume that the population variances are equal.

Using $\alpha = 0.05$, test whether the variance of the new machine is significantly smaller than the variance of the old machine. Assume both populations are normally distributed.

- State the null and alternative hypotheses.
(2 marks)
- Calculate the test statistic.
(2 marks)
- Find the critical value.
(4 marks)
- Test the claim and state the conclusion.
(2 marks)

QUESTION 3**(15 MARKS)**

A fitness center wants to compare the average weight loss achieved by participants following two different diet programmes after one month.

Two random samples are selected.

Diet Programme	Sample Size	Mean Weight Loss (kg)	Standard Deviation
Programme A	14	6.5	1.8
Programme B	12	5.2	1.5

Assume that the population variances are equal.

Using $\alpha = 0.05$, test whether Diet Programme A produces a significantly greater mean weight loss than Diet Programme B.

- a) State the null and alternative hypotheses.
(2 marks)
- b) Calculate the test statistic.
(7 marks)
- c) Find the critical value.
(3 marks)
- d) Test the claim and state the conclusion.
(3 marks)

QUESTION 4**(10 MARKS)**

A university librarian wants to analyse whether the number of books borrowed is uniformly distributed across the weekdays. This information is important for planning staff duty schedules, managing service counters, and improving overall library efficiency, especially during peak periods. By understanding borrowing patterns, the librarian can ensure sufficient resources are available and enhance user experience. To investigate this, the number of books borrowed each day was recorded over a randomly selected week during a regular academic semester, and the data are presented in the table below.

Day	Monday	Tuesday	Wednesday	Thursday	Friday
No. of books borrowed	42	38	45	35	50

- a) State the null and alternative hypotheses.

- (1 mark)
- b) Calculate the expected frequency and the test statistic.
- (5 marks)
- c) Find the critical value. Use significance level $\alpha = 0.05$.
- (2 marks)
- d) Test the claim and state the conclusion.
- (2 marks)

QUESTION 5

(15 MARKS)

An e-commerce company wants to understand whether online shopping frequency differs according to age group. This information is useful for designing marketing strategies, discount campaigns, and mobile app features. Younger and older customers may have different levels of confidence and habits when shopping online.

A random sample of 178 customers was selected. The data are shown below.

Online Shopping Frequency	Age 18 - 35	Age 36+
Rarely	25	27
Sometimes	30	28
Often	35	33

At $\alpha = 0.01$, test whether age group and online shopping frequency are related.

- a) State the null and alternative hypotheses.
- (2 marks)
- b) Calculate the expected frequency and the test statistic.
- (8 marks)
- c) Find the critical value.
- (3 marks)
- d) Test the claim and state the conclusion.
- (2 marks)